

Solution Manuel for Proposed Problems

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1 Tianlin

1.0.1 P1 Algebra

Solution:

Although this seems like a number theory problem at first, being labeled as algebra is a big giveaway.

We can first try to complete the square on the expression:

$$\begin{aligned} 2^{202} + 202 &= (2^{101})^2 + 2 \cdot 2^{101} + 1 - 2 \cdot 2^{101} + 201 \\ &= (2^{101} + 1)^2 - 2^{102} + 201 \end{aligned}$$

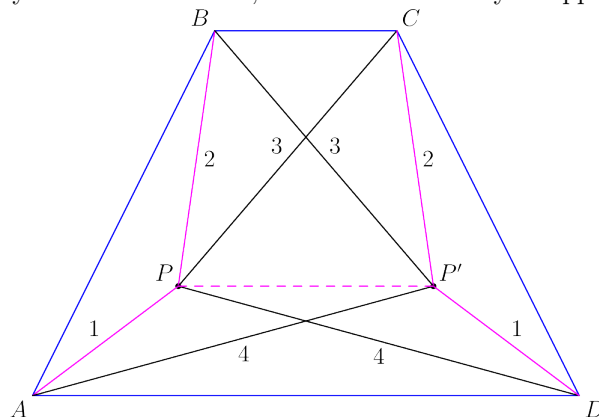
Then, we can apply the difference of squares on the expression, since both $(2^{101} + 1)^2$ and 2^{102} are squares:

$$(2^{101} + 1)^2 - 2^{102} + 201 = (2^{101} - 2^{51} + 1)(2^{101} + 2^{51} + 1) + 201.$$

Thus, we see that the remainder is 201

1.0.2 P2 Geometry

Solution: If we tried to brute force this problem, we would end up with very scary numbers. However, there is a better way to approach:



(diagram from AOPS)

Consider the reflection P' of P over the perpendicular bisector of $\overline{BC}$, creating two new isosceles trapezoids $DAPP'$ and $CBPP'$. Under this reflection, $P'A = PD = 4$, $P'D = PA = 1$, $P'C = PB = 2$, and $P'B = PC = 3$ (all labeled on the diagram).

Since $DAPP'$ and $CBPP'$ are isosceles trapezoids, they are cyclic. Using Ptolemy's theorem on $DAPP'$, we get that $(PP')(AD) + (PA)(P'D) = (AP')(PD)$, so

$$PP' \cdot AD + 1 \cdot 1 = 4 \cdot 4.$$

Then, using Ptolemy's theorem again on $CBPP'$, we get that $(BC)(PP') + (BP')(CP') = (BP')(CP)$, so

$$PP' \cdot BC + 2 \cdot 2 = 3 \cdot 3.$$

Thus, $PP' \cdot AD = 15$ and $PP' \cdot BC = 5$; dividing these two equations and taking the reciprocal yields $\frac{BC}{AD} = \frac{1}{3}$.

2 Franklin

2.0.1 P1 Combo

Fred must move up one row at a time. For each time he moves up, in a sense he can choose at which of the coordinates $x = 0, 1, 2, 3$, or 4 to move up on. This occurs from $y = 0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4$, so the answer is $5^4 = 625$

2.0.2 P2 Combo

Using geometric probability, the total volume is 125 cubic minutes. The good area is represented by a pyramid with base 12.5 squared minutes and height 5 minutes, which must have $\frac{1}{6}$ the total value. Thus, the probability is $\frac{1}{6}$.

3 Vincent

3.0.1 P1 Combo

Let the probability after the n th pass be $A(n)$. Then, if Alice gets the ball on the n th pass, it should not be in her hand on $n-1$. Also, after the $n-1$ th pass, if it is not in Alice's hand, there is a $\frac{1}{5}$ probability that it goes to her. Thus, the probability that it is in Alice's hand in the n th is $\frac{1}{5}(1-A(n-1))$. Then, $A(n) = \frac{1}{5}(1-A(n-1))$. Now, we can use this recursion to solve the problem. After 1 pass, the probability is 0. After the second, it is $\frac{1}{5}$, then $\frac{4}{25}$, then $\frac{21}{25 \cdot 5} = \frac{21}{125}$, then $\frac{104}{125 \cdot 5} = \frac{104}{625}$, then $\frac{521}{625 \cdot 5} = \frac{521}{3125}$. Finally, for the seventh one, the probability is $\frac{2604}{3125 \cdot 5} = \frac{2604}{15625}$.

3.0.2 P2 Combo

Bob should first go to the burger place, then his house, only moving in the positive direction. The gang is in the way between the burger place and his house. First, the burger place. To get there, we move in the x-direction 2 times, the y direction 1 time, and the z 3 times. Then, the amount of ways to get there is $6!/(2!1!3!) = 60$. Then, the amount of ways from the burger place to his house is $(4+5+3)!/(4!5!3!) = 27720$. However, the gang is in the way, and we should count the ways to go through the gang and subtract that. The total is $(2+2+2)!/2!2!2! * (2+3+1)!/2!3! = 90*60 = 5400$. Thus, the number of ways from the burger place to his house is $27720-5400 = 22320$. Therefore, the total number of ways to the burger place and then his home is $22320 * 60 = 1339200$